BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA40107
Course Name: Object Oriented Programming in JAVA
Weekly Contact Hours: 3L
Credits: 3
Total Allotted Hours: 45L

Course Objective: The aim of this course is to provide an introduction to object oriented programming (OOP) using the Java programming language. Its main objective is to teach the basic concepts and techniques which form the object oriented programming paradigm.

Pre-requisite(s): Knowledge of programming language.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Define the concept of object oriented design and compare OOPs with other programming languages.
CO2: Illustrate basic structure of Java program and apply different Java methods, constructs and libraries for real world applications.
CO3: Apply and analyze inheritances, interfaces and packages of Java.
CO4: Analyze and assess exception handling of Java.
CO5: Distinguish and develop threads and applet.

Module 1: Object Oriented Design **[4H]**

Concepts of object-oriented programming language, object, class, relationships among objects, relationships among classes-association, aggregation using instantiation, meta-class, difference between OOP and procedural language – advantages and disadvantages, concepts of message passing, overview of the properties of OOPs.

Module 2: Class and Object Properties **[16H]**

Basic concepts of java programming – advantages of java, byte-code & JVM, data types, access specifiers, operators, control statements and loops, array, creation of class, object, constructor, finalize and garbage collection, use of method overloading, this keyword, use of objects as parameter and methods returning objects, call by value and call by reference, static variables and methods, nested and inner classes, basic string handling concepts- string functions, concept of mutable and immutable string, command line arguments, basics of I/O operations – keyboard input using Buffered Reader and Scanner classes.

Module 3: Inheritance and Packages **[9H]**

Definition, types of inheritance, super class and subclasses including multilevel hierarchy, use of super and final keywords, dynamic method dispatch, use of abstract classes and methods, interfaces. creation of packages, importing packages, member access for packages.

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Module 4: Exception Handling **[6H]**

Exception handling basics, different types of exception classes, use of try and catch with throw, throws and finally, creation of user defined exception classes.

Module 5: Threads & Applet **[10H]**

Basics of multithreading, main thread, thread life cycle, creation of multiple threads, thread priorities, thread synchronization, inter-thread communication, deadlocks for threads, suspending and resuming threads, basics of applet programming, applet life cycle, event model and listener, creation of buttons & text fields.

Course Outcomes (COs) and Modules Mapping:

Sl. No	Course outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K2
2.	CO2	M2	K2, K3
3.	CO3	M3	K3, K4
4.	CO4	M4	K4, K5
5.	CO5	M5	K5, K6

Text Books:

1. Programming with Java: A Primer, E. Balagurusamy, TMH, 4th Edition, 2006.

Reference Books:

1. The complete reference-Java2, Patrick Naughton, Herbert Schildt, TMH, 5th Edition, 2017
2. Object Oriented System Development, Ali Bahrami, McGraw Hill, Classic Edition, 2017.